



— DISTRIBUTION RELIABILITY AND WINTER READINESS

Fewer outage hours, ready before *winter.*



One governed, bilingual view of the network that shows where outage hours concentrate, which transformers will overload next winter, and who is affected when a device fails.

AUDIENCE

Sponsor · Operations

STATUS

Working prototype

SCOPE

Assets · outages · peak

DATA

Synthetic · 3 CEDs



Outage hours concentrate
in a few zones, and *winter
overloads are predictable.*

What the prototype shows

Six months of outages and two winters of transformer load.

GRID GENIE · PROTOTYPE FINDINGS

59%

of customer-hours interrupted
came from 10% of protection zones

Reliability spend works hardest in a *small* part of the network.

Ranking protection zones by customer-hours interrupted (CHI) shows a steep concentration. Fourteen zones carry most of the outage burden, so vegetation, renewal and crew plans can be aimed instead of spread evenly.



TOP 5% OF ZONES	TOP 10% OF ZONES	ZONES WITH NO INTERRUPTION
41% of CHI	59% of CHI 14 zones	28% (38 of 138)

Three pressures on distribution, and what the data *already shows*.

PROTOTYPE ON A SYNTHETIC QUÉBEC NETWORK
3 CEDS · 23 LINES · 8,769 CUSTOMERS

01 THE PRESSURE

- Reliability, winter peak and ageing assets compete for one budget.
- Continuity indices such as SAIDI and SAIIFI are reported and scrutinised every year.
 - Electric heating pushes the system peak into the coldest days of January and February.
 - Many poles and transformers reach end of life over the same decade.

“The decision is where the next dollar removes the most outage hours.

FRAMING FOR THIS BRIEFING

02 WHAT THE PROTOTYPE FOUND

- The signal is already in GIS, outage and meter data.
- 59% of customer-hours interrupted sit in 10% of protection zones.
 - 110 transformers (8.7%) exceeded 150% of nameplate in winter 2026.
 - 32 of them had not overloaded the winter before, so a last-winter rule misses them.

“876 poles, one in five, were installed before 1975.

PROTOTYPE NETWORK · 4,355 POLES

03 WHAT CHANGES

- One governed view that operations can question in French or English.
- Every asset, span, zone and outage on one map, stored as native geometry.
 - A watch list of transformers likely to overload next winter, refreshed on every run.
 - Plain-language answers for dispatch: who is affected if this device fails?

“Customer names and exact locations stay masked unless a role requires them.

UNITY CATALOG GOVERNANCE

Measured on a working *prototype*.

01

IN THIS PART

Results · Genie in French ·
today compared with Grid
Genie

BUILT ON

One Databricks workspace and
one governance layer.

DATA

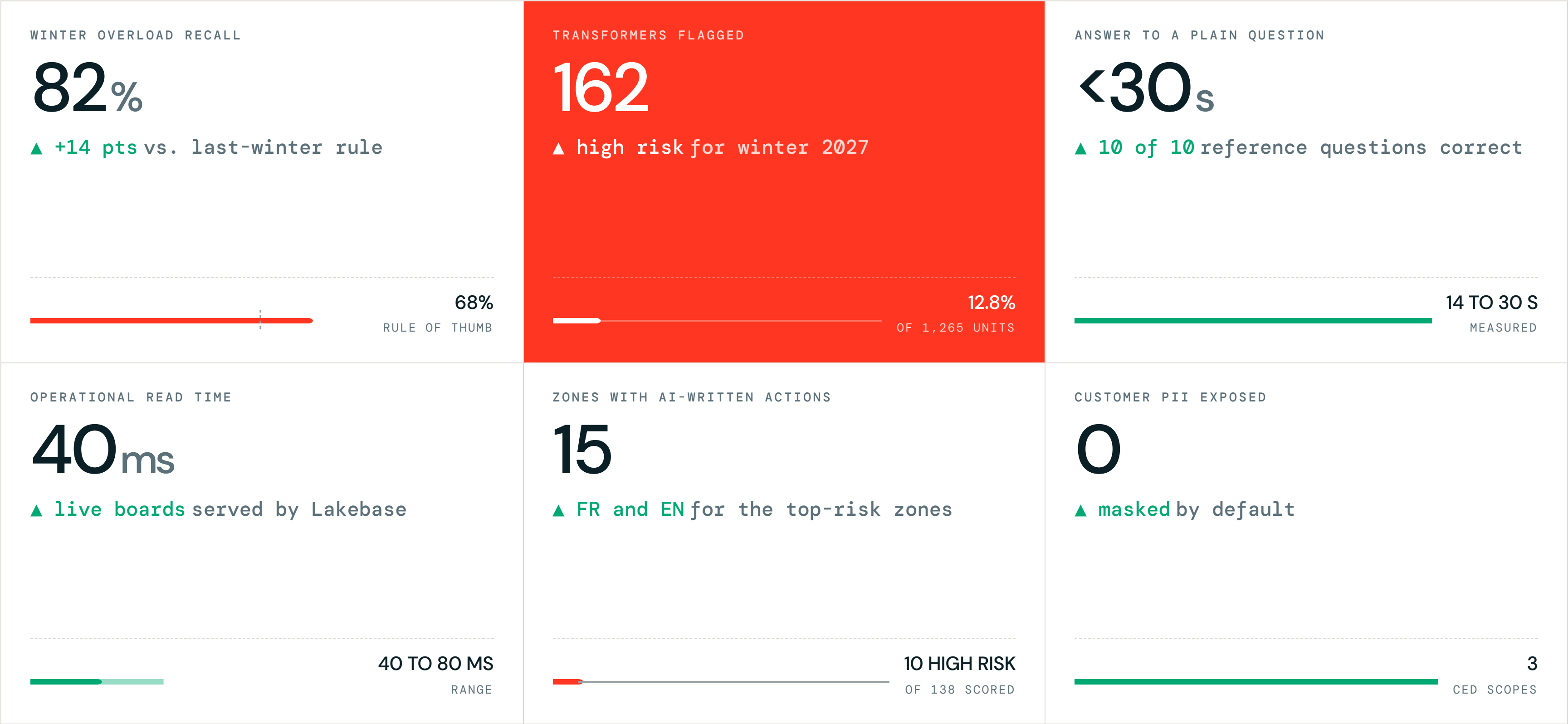
Synthetic network covering
Laval, Beauce and Matapédia.

MEASURED

Model accuracy, answer time
and read latency.

What the prototype *measured*

PROTOTYPE RUN · OCTOBER 2026
SYNTHETIC NETWORK · 3 CEDS



Dispatch asks in French. *Genie answers* in seconds.

Operators type the question they would send to a GIS analyst. Genie writes governed SQL with native spatial functions, returns the answer and draws the affected zone on the network map.

→ Bilingual.

Questions and answers in French or English, using utility vocabulary: poste, ligne, portée, coupe-circuit.

→ Spatial.

Distances, rights-of-way and zones are computed on real geometry, in metres.

→ Governed.

Answers follow the same masking and operating-centre scopes as every report.

→ On the map.

Every located answer appears on the network map for the dispatcher.

grid-genie-map / assistant Genie

DO · DISPATCH

DO

Combien de clients vont être affectés si le coupe-circuit LAV_Y3Z4G lâche ?

G

69 clients et 7 transformateurs sont en aval du coupe-circuit LAV_Y3Z4G, sur la ligne LAV_SVR_242. La zone est affichée sur la carte.

gold.protection_devices

EN AVAL · CHI 6 MOIS · TOP
LAV_Y3Z4G ZONES

Clients 69

Transformateurs 7

Ligne LAV_SVR_242

Type coupe-circuit

DO

Et quels transformateurs de cette ligne ont dépassé 150 % l'hiver dernier ?

Posez une question de suivi...

→

GRID GENIE - business briefing

Answer reproduced from the prototype Genie space

1 / 16

Operational questions answered in *seconds* instead of days.

TYPICAL CURRENT PRACTICE
TO BE VALIDATED IN DISCOVERY

QUESTION FROM THE BUSINESS	Grid Genie GOVERNED, ON ONE MAP	GIS request AD-HOC EXTRACT	Periodic reports BI AND SPREADSHEETS	Field experience LOCAL KNOWLEDGE
Customers affected if a protection device fails	■ Seconds, shown on the map	■ One to three days	■ Not available	■ Estimate
Zones driving customer-hours interrupted	■ Ranked, last 3 or 6 months	■ On request	■ Monthly, by region	■ Known hot spots
Transformers at risk next winter	■ Predicted, with a probability	■ Not modelled	■ Last winter's overloads	■ Known problem units
Oldest poles and longest spans on a line	■ One question	■ Extract on request	■ Not tracked	■ Partial
Customer data protected by role	■ Masked by default	■ Depends on the extract	■ Copies in circulation	■ Not applicable
Live operational status	■ Served in 40 to 80 ms	■ Not available	■ Next reporting cycle	■ Radio and phone
■ Answered directly ■ Partial or slow ■ Not answered today				

What it is *worth*, and to whom.

02

IN THIS PART

Two owners · value levers ·
illustrative value model

EXECUTIVE SPONSOR

Reliability indices, winter
readiness, capital
priorities.

DOMAIN OWNER

Dispatch answers, renewal
lists, crew planning.

NUMBERS

Measured on the prototype,
scaled with stated
assumptions.

Two owners read the same *governed* network data.

EXECUTIVE SPONSOR · VP DISTRIBUTION
DOMAIN OWNER · OPERATIONS AND ASSET MANAGEMENT

The sponsor gets a defensible link between reliability spend and outage hours. The domain owner gets answers in the time a dispatch decision takes.

For the executive sponsor, the question is where the next dollar of reliability and renewal budget removes the most customer-hours interrupted, and whether the network is ready for the January peak. Grid Genie ranks protection zones by CHI trend and asset age, and lists the transformers likely to overload next winter with a probability attached.

For distribution operations and asset management, the questions are practical: who is affected if this device fails, which poles on this line are oldest, where the longest spans are. Those answers used to wait for an analyst. In the prototype they arrive in under thirty seconds, in French or English, already drawn on the map.

Both owners read the same governed tables. Customer names and exact locations stay masked by default, and each operating centre sees its own territory, so self-service does not create new copies of sensitive data.

01 SPONSOR · RELIABILITY

-12% SAIDI

Illustrative effect of a 20% CHI cut in the top 10% of zones.

02 SPONSOR · WINTER READINESS

+21% found

More at-risk transformers identified before the peak than with a last-winter rule.

03 OWNER · DISPATCH

<30 s

Customers affected by a device, answered from a plain French question.

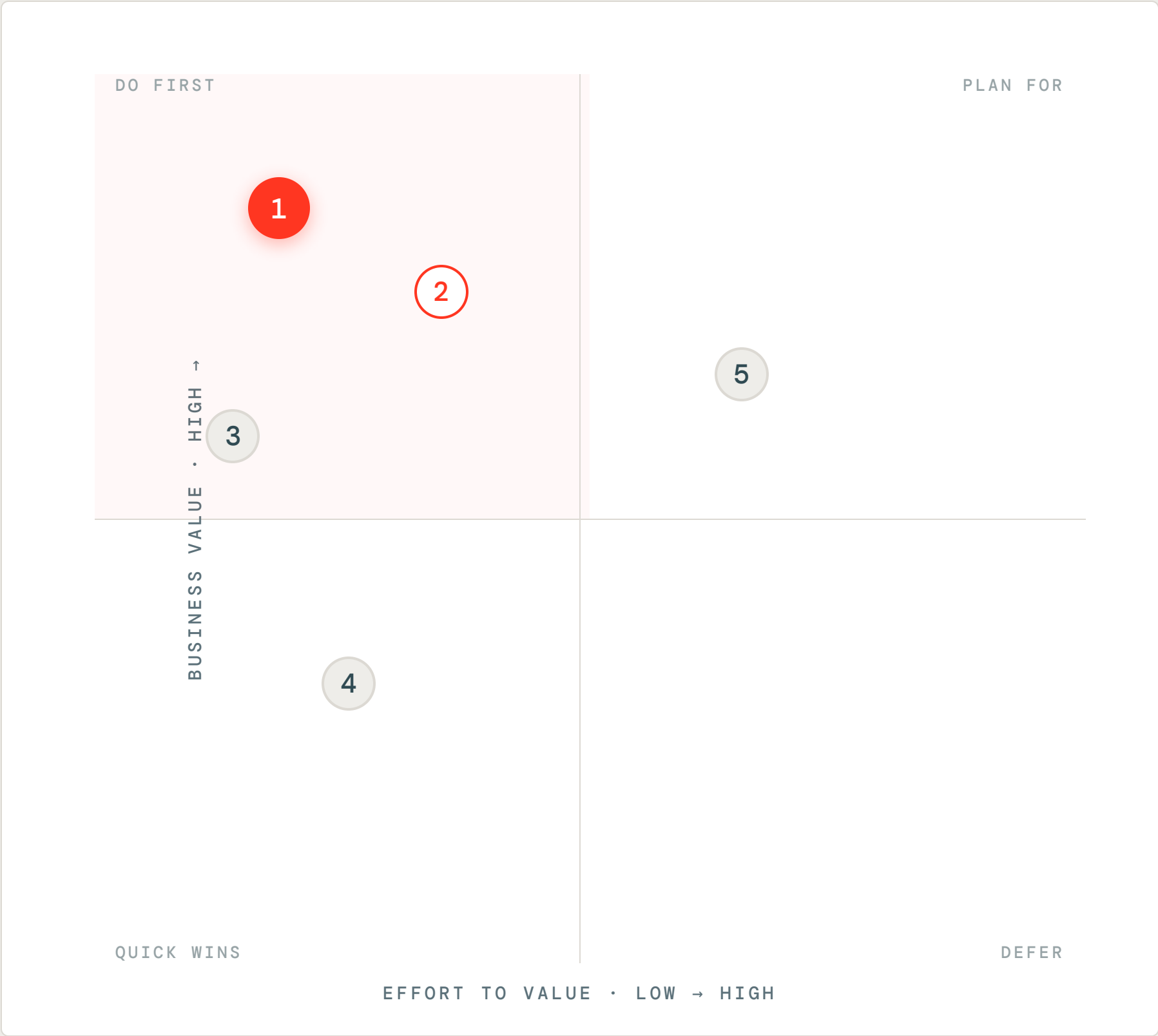
04 OWNER · ASSET PLANNING

162 units

Ranked transformer watch list for winter 2027 on the prototype network.

Five value levers, *ranked* by impact and effort.

HYPOTHESIS TO PRESSURE-TEST
WITH DISTRIBUTION OPERATIONS



WHERE WE WOULD START, IN ORDER

- 1 Winter overload watch list**
Asset management · act before December. **+21%**
at-risk units found
- 2 CHI hot-spot targeting**
Reliability programme · aim vegetation and renewal. **-12%**
SAIDI, illustrative
- 3 Outage-impact answers for dispatch**
Operations · plain-language questions on the map. **<30 s**
time to answer
- 4 Live operational boards**
Control room · status served by Lakebase. **40 to 80 ms**
read time
- 5 Self-service for every CED**
All operating centres · governed, no data copies. **3 CEDs**
one platform

An illustrative value model for *one million* customers.

PROTOTYPE RATIOS × STATED ASSUMPTIONS
REPLACE WITH THE UTILITY'S BASELINES IN THE PILOT

KPI BASIS OF THE ESTIMATE	TODAY BASELINE	WITH GRID GENIE ESTIMATE	CHANGE PER YEAR
System customer-hours interrupted (SAIDI) 59% of CHI in 10% of zones × an assumed 20% reduction in those zones	Index 100	Index 88	-12%
At-risk transformers found before winter Measured holdout recall: 82% for the model, 68% for a last-winter rule	68%	82%	+21% relative
Emergency winter transformer failures Assumed: 143k units, 8% overloaded, 3% of missed units fail; misses drop from 32% to 18%	about 110	about 62	-48 units
Time to answer a spatial operations question Measured 14 to 30 s; today assumed one to three days per analyst request	1 to 3 days	<30 s	>99% faster
Copies of customer data made for analysis Masking and operating-centre scopes enforced in Unity Catalog	Per request	None	Removed

■ Estimate with Grid Genie

□ Headline lever

 Illustrative only · every assumption is listed in its row · calibrate during the pilot

Twelve weeks to a *winter-ready* operating centre.



IN THIS PART

Platform · pilot plan · next step

SCOPE

One CED, two winters of history, real GIS and outage data.

TEAM

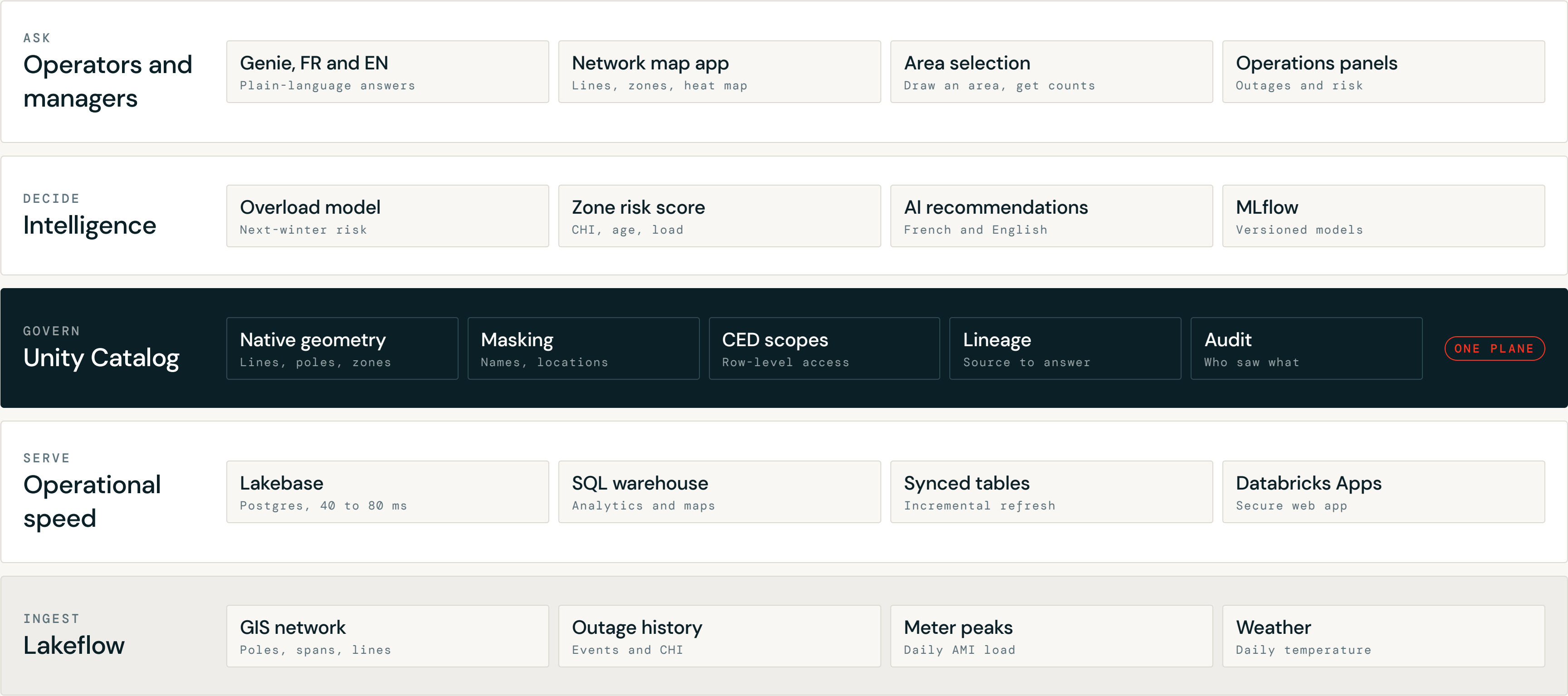
Distribution operations, asset management, IT data team, Databricks.

OUTPUT

A measured readout against the utility's own baselines.

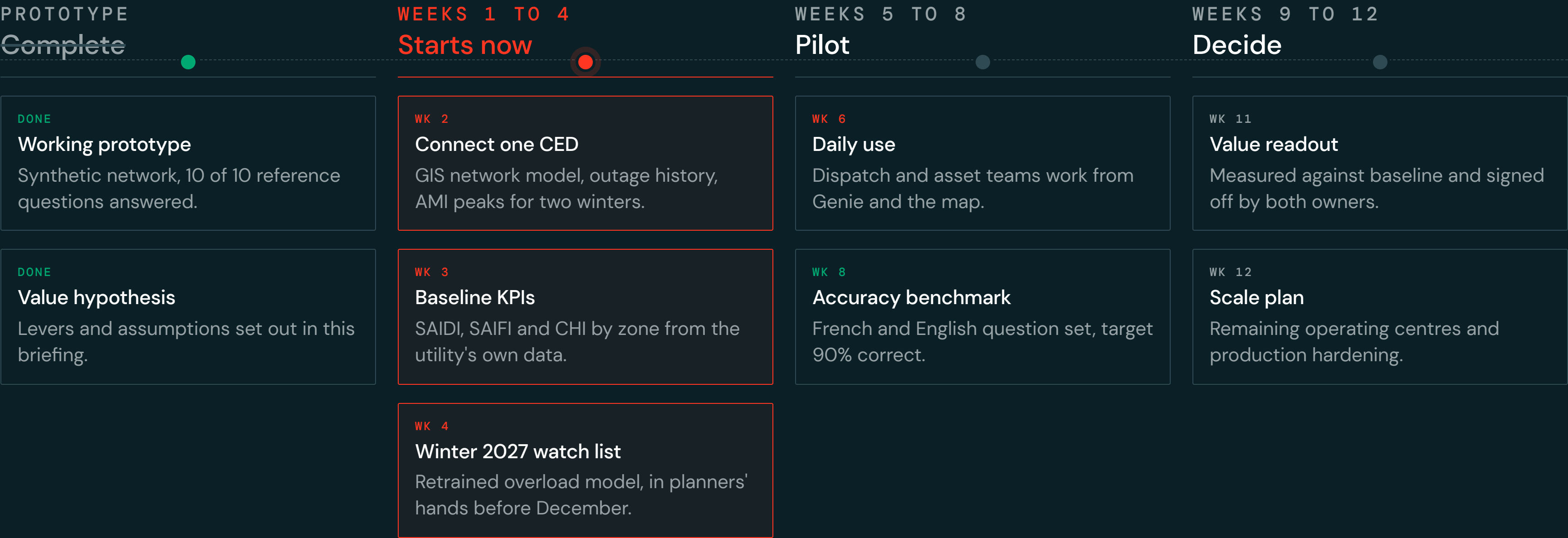
One governed platform from meter data to the *dispatcher*.

DATABRICKS DATA INTELLIGENCE PLATFORM
RUNNING END TO END IN THE PROTOTYPE



A 12-week pilot with *one* operating centre.

PROPOSED PLAN · START OCTOBER 2026
WATCH LIST DELIVERED BEFORE DECEMBER



● Done ● Starts now ● Planned

Each phase ends with a joint checkpoint

— PROPOSED NEXT STEP

Pilot one operating centre *before the peak.*

DECISION NEEDED

An executive sponsor and one CED



01 Name the sponsor and the CED

One executive owner, one operations lead, one asset lead.

02 Share two winters of data

GIS network model, outage history and AMI daily peaks.

03 Run the 12-week pilot

Watch list before December, measured readout in week 12.

LIVE DEMO

Map app and Genie space

CONTACT

Your Databricks account team